

CTGAD-X: Explainable Contrastive Learning on Temporal Graphs for Advanced Threat Detection

Hafiz Bilal Ahmad , Haichang Gao , *Member, IEEE*, Naila Latif , and Ping Wang 

Abstract—Advanced persistent threats (APTs) and zero-day attacks evade signature-driven and fully supervised intrusion detection, motivating the development of anomaly detectors that generalize to unseen behaviors while remaining interpretable in high-stakes operations. Existing temporal graph neural network (GNN) detectors often rely on reconstruction objectives and offer limited transparency, which can weaken the separability of novel threats and reduce analyst trust. We propose CTGAD-X, an explainable self-supervised framework for zero-day threat detection on temporal graphs. CTGAD-X learns a discriminative normal embedding space from benign traffic using contrastive learning and flags anomalies based on distance-based deviation from a normal reference set. To ensure temporal correctness, we introduced causality-aware overlap-based flow slicing and validated window-level attribution against raw PCAP packet timestamps, achieving strong agreement (Pearson $r = 0.943$; median relative error 3.2% for bytes and 2.0% for packets). On UNSW-NB15 under causality-aware slicing, CTGAD-X achieves ROC-AUC = 0.9908 and F1-score = 0.9888. In the zero-day leave-one-out evaluation, held-out *Exploits* increase from $\approx 88.8\%$ detection at a deployable low-false-alarm (fixed-threshold) point to 100% at a higher sensitivity. Similarly, *Generic* increases from $\approx 65.5\%$ to 100%, quantifying the recall-false-alarm trade-off. To support operational transparency, CTGAD-X integrates a multi-level XAI module (attention attribution, input perturbation, and global permutation importance) and validates explanation via feature-masking and perturbation tests. We further evaluated cross-dataset generalization on multiple benchmark datasets using a transfer learning with minimal adaptation. CTGAD-X is lightweight (16,320 parameters) and achieves 3.94 ms encoder inference per sequence, enabling practical deployment in high-throughput environments. Finally, CTGAD-X unifies temporally consistent graph construction, contrastive representation learning, and reliable explanations to enable reliable zero-day detection.

Index Terms—Anomaly detection, zero-day attack detection, temporal graphs, graph neural networks, contrastive learning, explainable AI (XAI), network security.

I. INTRODUCTION

THE escalation of advanced persistent threats (APTs) and zero-day attacks represents a paradigm shift in

the cybersecurity landscape, challenging the foundational assumptions of conventional defense mechanisms. High-profile incidents, such as the StuxNet malware targeting critical infrastructure and the SolarWinds supply chain attack, have demonstrated how stealthy multi-stage intrusions can bypass perimeter defenses and persist undetected for months, leading to catastrophic data breaches and systemic compromises [1].

Characterized by their customized nature and exploitation of previously unknown vulnerabilities, these threats cause signature-based detection to be entirely ineffective by definition. The increasing strategic and economic costs of such breaches underscore the critical and urgent need for a new generation of Network Intrusion Detection Systems (NIDS) capable of proactive, behavior-centric threat identification.

Despite significant progress, a critical gap persists between the capabilities of existing NIDS and the demand to detect modern threats. Although signature-based systems are inherently reactive, even adaptive machine learning models often fail to capture the relational and temporal context of network communications by treating flows as independent and identically distributed (i.i.d.) events, which contributes to high false positive rates (FPR) [2].

Graph-based representations have emerged as powerful alternatives that capture the complex web of interactions within a network. However, early approaches that rely on static graphs provide only a time-aggregated snapshot, failing to model the evolutionary dynamics crucial for detecting multistage APT campaigns [3]. Recent developments have employed temporal Graph Neural Networks (GNNs); however, many are predicated on reconstruction or prediction-based learning objectives. A fundamental and frequently neglected limitation of this approach is that a novel attack sequence, when exhibiting sufficiently plausible local structural patterns, may fail to produce a significant reconstruction error, thereby allowing it to evade detection [4]. Furthermore, the increasing complexity of these deep learning models has worsened their “black-box” nature, creating a significant barrier to trust and operational adoption in security environments where interpretability is paramount for effective incident response [5].

Beyond the choice of learning objectives and the need for interpretability, flow-summary datasets can suffer from temporal leakage owing to windowing. When a flow spans multiple windows, assigning its full counters, such as bytes and packets, to the start window can introduce future traffic into earlier snapshots and bias the resulting temporal graphs. This leakage can inflate the reported detection performance and distort the learned temporal dynamics and associated explanations.

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To bridge these gaps, this study introduces **CTGAD-X**, an innovative framework for Contrastive Temporal Graph Anomaly Detection with eXplainability, which combines temporally consistent graph construction, self-supervised contrastive representation learning, and multi-level explanations. Instead of learning to reconstruct network traffic, CTGAD-X uses a contrastive learning paradigm to learn a highly discriminative embedding space. Inside this space, normal network sequences are mapped into a tight cluster, whereas anomalous sequences are pushed far away. This objective encourages the model to capture the global properties of the normal dynamic behavior, making it sensitive to deviations that may correspond to unseen attacks [6], [7]. For operational transparency, CTGAD-X provides complementary explanations by combining attention-based attribution with input perturbation, enabling analysts to identify why a specific alert is triggered [8].

The primary contributions of this study are fourfold.

- 1) *Causality-Aware Temporal Graph Construction with Packet-Level Validation*: We mitigate look-ahead bias in flow-summary windowing by introducing overlap-based flow slicing, which allocates flow statistics across all intersecting windows (instead of start-time-only attribution), enforcing temporal causality. We validated the resulting window-level aggregates against packet-timestamp references from raw PCAP traces, obtaining a strong agreement (Pearson $r = 0.943$).
- 2) *Normal-Only Contrastive Manifold Learning for Zero-Day Detection*: We propose CTGAD-X, a self-supervised contrastive framework that trains a Siamese spatio-temporal encoder exclusively on normal traffic sequences to learn a compact embedding manifold. Anomalies are detected as outliers based on the distance to a normal reference bank in the learned representation space.
- 3) *Multi-Level Explainability with Quantitative Evaluation*: We provide complementary explanations at three levels: (i) structural and temporal attention to localize influential nodes/edges and time steps, (ii) input perturbation to measure each component's counterfactual impact on the anomaly score, and (iii) global permutation importance to summarize dataset-level feature reliance. We evaluated the explanation quality using Fidelity⁺ (faithfulness) and Stability (robustness).
- 4) *Rigorous Zero-Day and Cross-Dataset Evaluation Protocols*: We introduce a systematic held-out attack-family (leave-one-out) protocol on UNSW-NB15 [9] to assess the true zero-day behavior for specific unseen threat families (e.g., *Exploits*, *Generic*). We further evaluate transfer across heterogeneous benchmarks (CICIDS-2017 [10], ToN-IoT [11], SWaT [12], and SCVIC-APT [13]) under consistent training assumptions (normal-only fine-tuning), demonstrating strong generalization with limited adaptation.

II. RELATED WORK

The detection of sophisticated cyber threats, particularly Advanced Persistent Threats (APTs) and zero-day attacks,

continues to be a central concern in cybersecurity research. This section reviews the key literature on network intrusion detection, graph-based anomaly detection, and explainable AI (XAI) to contextualize the proposed CTGAD-X framework.

Early Network Intrusion Detection Systems (NIDS) evolved from signature-based methods to statistical and machine learning (ML) approaches. However, these methods often suffer from high false positive rates (FPR) and rely on extensive labeled data. Critically, most traditional methods treat network events as independent, failing to capture the complex relational and temporal patterns inherent to sophisticated attacks [14].

Representing network traffic as graphs captures the interactions of complex entities. Early methods focused on static snapshots; however, APTs are inherently dynamic, necessitating temporal graph analysis. However, traditional temporal approaches often rely on hand-crafted features, limiting their ability to detect novel threats.

Recent studies have demonstrated that GNNs, including GCNs and GATs, can effectively model dependencies in static network data [15], [16]. To address dynamic environments, temporal GNNs (e.g., EvolveGCN and DySAT) learn representations from graph sequences. However, many rely on reconstruction- or prediction-based objectives, which may struggle to detect novel attacks with patterns resembling normal traffic [17].

The recent literature has expanded the zero-day defense landscape. A systematic review by Yee et al. [18] identified adaptability as a key challenge. To address generalization to unseen data, Dai et al. [19] proposed a machine learning framework for zero-day detection, whereas Sana et al. [20] explored Vision Transformers for robust representation. Despite these advances, many state-of-the-art graph frameworks, such as ANUBIS [21], rely on supervised or label-dependent training, limiting their applicability to novel zero-day scenarios.

Beyond general zero-day detection, several studies have examined the robustness against adaptive attackers and APT-like behaviors in operational settings, such as critical infrastructure. In our prior work [22], we investigated adversarial machine learning considerations for advanced threat detection inspired by Stuxnet-style campaigns, highlighting how attacker adaptation can distort the learned signatures and evaluation outcomes. In contrast, CTGAD-X focuses on *normal-only* contrastive representation learning on *temporal graphs* with explicit causality-corrected windowing and multi-level explainability, targeting deployable zero-day generalization rather than supervised signature refinement.

Contrastive learning enables robust representation learning without labels [7]. Recent studies, such as PPT-GNN [23], have applied contrastive pre-training to graph-based intrusion detection but have not explicitly addressed unsupervised zero-day generalization. Similarly, self-supervised approaches, such as Anomal-E [24] and CAGN-GAT [25] demonstrate the power of representation learning but lack temporal modeling, explainability, or explicit zero-day evaluation protocols (see Table I).

Table I summarizes how CTGAD-X differs from representative paradigms along dimensions relevant to zero-day

TABLE I
COMPARATIVE INNOVATION MATRIX: CTGAD-X VERSUS REPRESENTATIVE PARADIGMS FOR TEMPORAL GRAPH INTRUSION/ANOMALY DETECTION

Work / Paradigm	Learning objective and decision rule	Temporal	Graph	Contrastive	Recon.	Zero-day protocol	XAI
Reconstruction-based temporal GNN-AE (e.g., TGAE-like [26])	Reconstruct/predict snapshots; anomaly via reconstruction error threshold	✓	✓	✗	✓	✗	✗
Unsupervised temporal link-prediction (e.g., DySAT [27])	Predict temporal edges/links; anomaly via prediction deviation / score	✓	✓	✗	✗	✗	✗
Self-supervised graph intrusion (e.g., Anomal-E [24])	Self-supervised graph representation (DGI-based); anomaly scoring varies	✗	✓	✓	✗	✗	✗
Temporal Contrastive IDS (e.g., TCG-IDS [28])	Multi-view temporal contrastive learning; anomaly via similarity score deviation	✓	✓	✓	✗	✗	✗
Bayesian/supervised provenance-graph APT (e.g., ANUBIS [21])	Supervised with calibrated uncertainty; decision via posterior / classifier	✓	✓	✗	✗	✗	✓
CTGAD-X (Proposed)	Normal-only NT-Xent to learn a discriminative normal manifold; anomaly via k-NN	✓	✓	✓	✗	✓	✓

Notes: “Zero-day protocol” indicates an explicit held-out (leave-one-out) evaluation where an attack family is unseen during training. This matrix summarizes methodological properties, not just performance.

deployment: training supervision, embedding geometry, evaluation protocol, and explainability of the model.

Explainability (XAI) is essential for the practical deployment of AI-based security systems [29]. Although general XAI methods exist, there remains a gap in techniques tailored to interpret anomaly scores from contrastively learned temporal graph embeddings. CTGAD-X addresses the threefold gap identified in prior work by integrating: (1) a temporal GNN encoder for dynamic pattern modeling, (2) contrastive learning on normal data for zero-day detection, and (3) a multi-level explainability module for actionable interpretations.

III. THREAT MODEL AND PROBLEM FORMULATION

Detecting sophisticated cyber threats, particularly Advanced Persistent Threats (APTs) and zero-day attacks, necessitates a clear understanding of the adversary’s behavior and the specific challenges they pose. This section outlines the threat model underpinning our work and formally defines the problem addressed by the CTGAD-X.

A. Adversary Profile and Capabilities

We consider a sophisticated adversary characteristic of APT actors conducting stealthy, “low-and-slow” operations over extended periods. This adversary may leverage zero-day exploits and dynamic TTPs, often following a multi-stage attack lifecycle (e.g., reconnaissance, C2, and lateral movement) designed to blend with normal traffic and evade traditional detection [18].

The primary challenge lies in identifying these evolving, potentially novel malicious activities among a high volume of benign network traffic.

B. Attack Manifestations in Network Traffic

In network telemetry, these TTPs manifest as anomalous sequences of interactions, such as unusual scanning for reconnaissance, the establishment of covert C2 channels using non-standard protocols, atypical internal lateral movement between segments, and patterned data exfiltration to external destinations.

C. Problem Formulation

Given a sequence of network interactions represented as a dynamic graph $\mathcal{G} = \{G_1, G_2, \dots, G_T\}$, where each snapshot $G_t = (V_t, E_t, X_t)$ captures the network state (nodes V_t , edges E_t , and node features X_t) within a discrete time window t , the problem is to:

- 1) *Learn Normal Behavior*: Develop a model M that learns a representation of “normal” or “expected” dynamic graph behavior from a training dataset $\mathcal{G}_{train}$. $\mathcal{G}_{train}$ primarily consists of normal traffic.
- 2) *Detect Anomalous Sequences (Zero-Day Focus)*: For an arbitrary input sequence of graph snapshots $\mathcal{S}_{test} = \{G_k, G_{k+1}, \dots, G_{k+L-1}\}$, where L is the sequence length, assign an anomaly score $S_{anomaly}(\mathcal{S}_{test})$. This score should be significantly higher if $\mathcal{S}_{test}$ contains patterns indicative of the threat model (Section III-B), including patterns corresponding to attacks not observed during the training of M (zero-day attacks).
- 3) *Provide Explainability*: For sequences flagged as anomalous, provide insights into *which* structural components (nodes, edges), temporal aspects (time steps), or features within $\mathcal{S}_{test}$ contributed most to its high anomaly score.

The primary objective is to maximize the detection rate of true anomalies (especially zero-days) while minimizing false alarms and ensuring that the detection process is interpretable for security analysts [29]. The proposed CTGAD-X framework directly addresses these objectives.

IV. METHODOLOGY

To address the challenge of detecting sophisticated and zero-day threats in dynamic network environments, we propose CTGAD-X, an explainable anomaly detection framework. CTGAD-X is designed to learn robust representations of network behavior by leveraging contrastive learning on temporal graph structures. By modeling expected communication patterns, it can identify significant deviations indicative of potentially unseen malicious activities. A key component of the

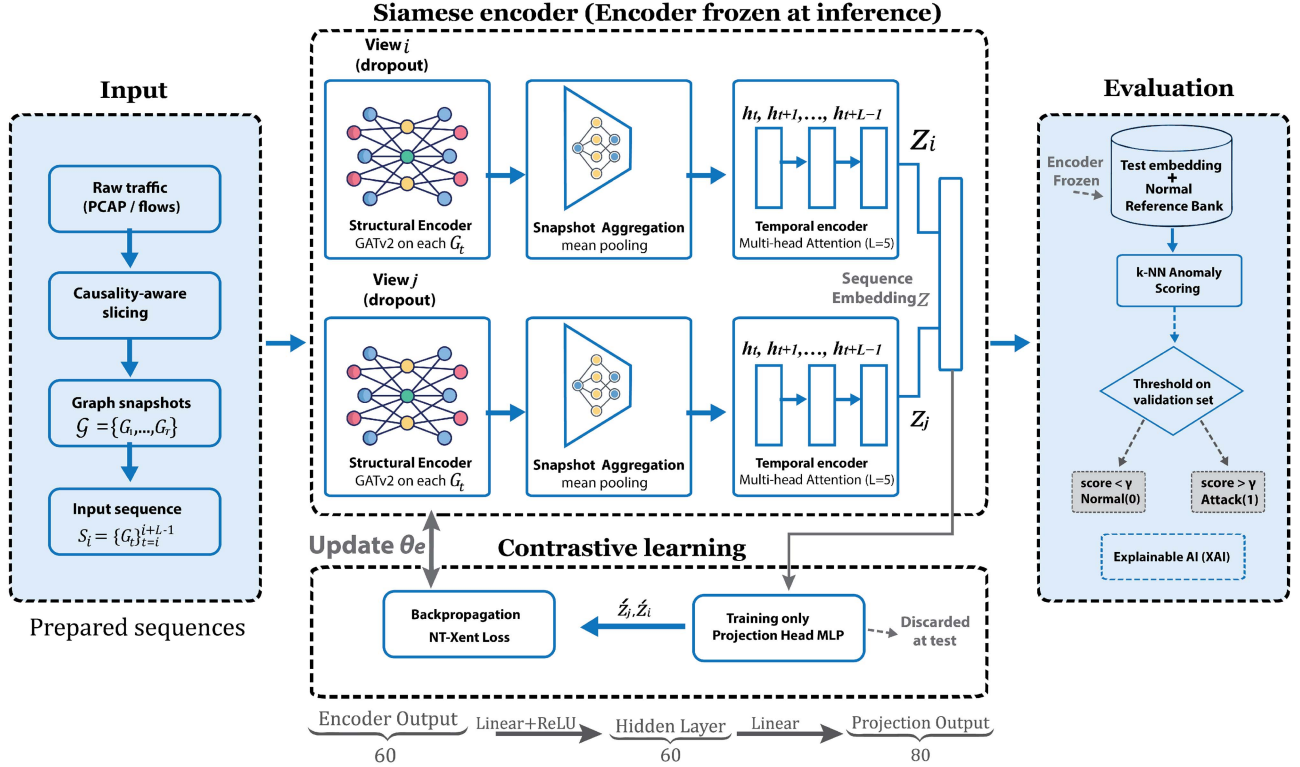


Fig. 1. Overview of CTGAD-X. (Left) Raw traffic (PCAP/flows) is converted into temporal graph sequences via causality-aware overlap-based slicing. (Center) A Siamese spatio-temporal encoder (shared weights, stochastic dropout views) is trained on normal sequences using the NT-Xent objective to learn a compact normal embedding space ($D_{\text{enc}} = 60$); the projection head ($D_{\text{proj}} = 80$) is used only during training and discarded at test time. (Right) With the encoder frozen, a test sequence embedding is scored by k-NN cosine distance to a normal reference bank for anomaly detection; post-hoc explanations are produced by the multi-level XAI module.

framework is its integrated explainability module, which provides critical insights into the factors that drive an anomaly score. The overall pipeline encompasses dynamic graph construction, temporal graph encoding, contrastive learning, anomaly scoring, and multi-faceted explainability analysis, as illustrated in Fig. 1.

A. Temporal Graph Formulation and Dataset Construction

We model network traffic from the UNSW-NB15 dataset [9] as a sequence of discrete-time graph snapshots $\mathcal{G} = \{G_1, \dots, G_T\}$. Each snapshot $G_t = (V, E_t, \mathbf{X}_t)$ captures the network state within a fixed window of duration Δt (default 30 s), where V is the static set of all observed IPs, E_t is the directed edge set, and $\mathbf{X}_t \in \mathbb{R}^{|V| \times d}$ is the node feature matrix.

1) *Causality-Aware Flow-to-Window Attribution*: The standard start-time-only windowing assigns the entire flow statistics to the start window, causing long-duration flows to inject future traffic into earlier snapshots. To eliminate this start-time aggregation artifact and mitigate temporal leakage, we implemented *overlap-based slicing*: each flow interval $[t_s, t_e)$ contributes to all intersecting windows proportionally.

For window $W_m = [w_m, w_m + \Delta t)$, the overlap with the flow is:

$$\Delta_m = \max(0, \min(t_e, w_m + \Delta t) - \max(t_s, w_m)). \quad (1)$$

Only windows with $\Delta_m > 0$ receive contributions. The feature allocation follows a uniform-rate approximation:

- **Additive counters** (bytes, packets): allocated as $x_m = x \cdot \Delta_m / \text{dur}$.
- **Rate features** (load): recomputed per slice from allocated counters.
- **State features** (TTL): aggregated via mean pooling (sum/count).

Although this assumes that traffic is distributed uniformly within a flow, our validation confirms that this is appropriate for the 30 s granularity used. The procedure is formalized in Algorithm 1.

2) *Ground-Truth PCAP Validation*: To verify that the uniform-rate assumption yields accurate window-level aggregation, we cross-validated the CSV-derived statistics against the ground-truth packet timestamps from the raw PCAP files (processed via *tshark*). We restricted the analysis to 159 matched windows representing the intersection of CSV and PCAP coverage. Comparing overlap-sliced aggregates against this packet-level reference yielded strong agreement: Pearson $r = 0.943$, with Median Relative Error = 3.2% (bytes) and 2.0% (packets) Table IV (Panel B), confirming that the method accurately reconstructs temporal profiles without look-ahead bias.

a) *PCAP/CSV Coverage Alignment*: The raw PCAP subset used for validation spans 194 fixed-length windows (at $\Delta = 30$ s)

Algorithm 1: Causality-Aware Overlap-Based Flow Slicing.

Input: Flow records $\{(u, v, t_s, \text{dur}, \mathbf{f})\}$, window size Δt

```

1: Initialize snapshots  $\{G_m\}$  with windows
    $W_m = [w_m, w_m + \Delta t)$ 
2: for all flow  $(u, v, t_s, \text{dur}, \mathbf{f})$  do
3:    $t_e \leftarrow t_s + \text{dur}$ 
4:   for all windows  $W_m$  intersecting  $[t_s, t_e)$  do
5:     Compute  $\Delta_m$  via (1)
6:     if  $\Delta_m > 0$  then
7:       Allocate features proportionally; add edge  $(u, v)$ 
       to  $E_m$ 
8:     end if
9:   end for
10: end for

```

TABLE II
DATASET PARTITIONING STRATEGY FOR MODEL TRAINING, EVALUATION, AND
ZERO-DAY GENERALIZATION ANALYSIS

Phase / Set	Component / Category	# Sequences	Total
I. Main Experiment			
Training Set	Normal	895	895
Validation Set	Unseen Normal	127	127
Test Set	Unseen Normal	251	1,319
	Unseen Attack	1,068	
II. Zero-Day Generalization (Held-Out Evaluation)			
Zero-Day	Exploits Class	1,591	3,111
Attack Sets	Generic Class	1,520	

TABLE III
KEY HYPERPARAMETERS FOR THE CTGAD-X FRAMEWORK (OPTIMIZED VIA
BAYESIAN SEARCH)

Hyperparameter Group	Value
<i>Model Architecture</i>	
Sequence Length (SEQ_LEN)	5
Encoder Layers (Structural / Temporal)	1 / 1
Encoder Dimension (STRUCTURAL_OUT_DIM)	60
Attention Heads (STRUCTURAL/TEMPORAL)	3
Projection Head Structure	MLP (60 → 60 → 80)
Max. Training Epochs (NUM_EPOCHS)	200
Early Stopping Patience	10
Batch Size (BATCH_SIZE)	64
Learning Rate (LEARNING_RATE)	1×10^{-3}
Weight Decay (WEIGHT_DECAY)	1×10^{-5}
Dropout Rate (DROPOUT)	0.5
Contrastive Temperature (τ)	0.1
k for k-NN Score (KNN_K)	15

based on packet timestamps, whereas the corresponding flow-summary CSV records cover only 159 windows after timestamp alignment and basic sanity filtering (e.g., valid start time and non-negative duration for overlap slicing). Consequently, there are 35 *PCAP-only* windows for which no corresponding CSV aggregates can be formed. We therefore compute agreement metrics (Pearson r , Spearman ρ , and median relative error) only

over the 159 matched windows, i.e., the intersection of PCAP and CSV coverage.

3) *Node Features and Data Partitioning*: For each node v in window t , we aggregate sliced statistics into an 11-dimensional feature vector $\mathbf{X}_t[v]$: traffic volume (sbytes, dbytes, Spkts, Dpkts), loss indicators (sloss, dloss), load rates (Sload, Dload), TTL statistics (sttl, dttl), and total duration (dur). To prevent leakage, we partitioned the chronologically sorted normal data into training (70%), validation (10%), and test (20%) sets. Feature scaling (StandardScaler) was fitted exclusively on the training split. Details are in Table II.

B. The CTGAD-X Framework: Architecture and Learning

The core of our methodology is the Contrastive Temporal Graph Anomaly Detection (CTGAD) framework, which is designed to learn discriminative embeddings from dynamic network traffic. The framework takes as input a dynamic graph sequence $\mathcal{G} = \{G_1, G_2, \dots, G_T\}$, where each snapshot $G_t = (V_t, E_t, X_t)$ represents the state of the network at time t with node features $X_t \in \mathbb{R}^{N \times F}$ and adjacency information E_t .

Its purpose is to learn discriminative embeddings $z_S \in \mathbb{R}^{D_{\text{enc}}}$ for input graph sequences $\mathcal{S}_i = \{G_t\}_{t=i}^{i+L-1}$ of length L . These embeddings are learned such that normal sequences reside in a compact region of the embedding space, enabling the detection of anomalous sequences, including unseen zero-day attacks, as outliers based on their distance from this normal manifold.

The framework comprises a specialized temporal GNN encoder, a contrastive learning objective for training, and a k-NN-based mechanism for anomaly scoring.

1) *Temporal Graph Encoder Architecture* (CTGAD_Encoder): The encoder is designed to capture both the spatial structure intra-snapshot and the temporal dynamics between snapshots within a sequence $\mathcal{S}_i$. It consists of stacked layers that progressively refine the node representations as follows:

- 1) *Structural Attention Layer*: To capture relational structure within each snapshot G_t , we use a Graph Attention Network v2 (GATv2) layer. For each node v , its initial features are $h_v^{(t,0)} = X_t[v]$. We update the node representations via multi-head attention over the neighborhood $\mathcal{N}(v)$:

$$e_{vu}^{(t,k)} = \mathbf{a}_k^\top \sigma \left(\mathbf{W}_k \left[h_v^{(t,0)} \parallel h_u^{(t,0)} \right] \right),$$

$$\alpha_{vu}^{(t,k)} = \frac{\exp(e_{vu}^{(t,k)})}{\sum_{u' \in \mathcal{N}(v)} \exp(e_{vu'}^{(t,k)})}. \quad (2)$$

and updates node embeddings by

$$h_v^{(t,1,k)} = \sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(t,k)} \mathbf{W}_k h_u^{(t,0)},$$

$$h_v^{(t,1)} = \bigparallel_{k=1}^K h_v^{(t,1,k)}. \quad (3)$$

followed by ELU activation. This is applied independently to each snapshot, yielding spatially-aware embeddings $\{H_t^{(1)}\}_{t=i}^{i+L-1}$ with $H_t^{(1)} \in \mathbb{R}^{N \times D_{\text{enc}}}$.

TABLE IV

CTGAD-X DETECTION PERFORMANCE UNDER CAUSALITY-AWARE PREPROCESSING AND EMPIRICAL VALIDATION OF TEMPORAL ATTRIBUTION AGAINST PCAP GROUND TRUTH. PANEL (A) REPORTS DETECTION METRICS ACROSS WINDOW SIZES; PANEL (B) CONFIRMS THAT CSV-DERIVED WINDOW AGGREGATES STRONGLY CORRELATE WITH PACKET-TIMESTAMP REFERENCE VALUES.

A. Detection Performance (Causality-Aware Slicing)						
Window	Accuracy	F1	ROC-AUC	AUPRC	FPR	Optimal k
30s (original)*	0.9986	0.9863	0.9986	0.9998	0.0159	15
30s (causal)	0.9909	0.9888	0.9908	0.9979	0.0204	15
60s (causal)	0.9881	0.9872	0.9786	0.9779	0.0501	15
120s (causal)	0.9740	0.9729	0.9975	0.9996	0.0175	15
*Pre-revision baseline using start-time-only window assignment (included for comparison).						
B. PCAP Validation (30s windows, 3 files, 10.5M packets)						
Metric	Pearson r	Spearman ρ	Median Rel. Err.	Matched	PCAP / CSV	
Bytes/window	0.943	0.952	3.2%	159	194 / 159	
Packets/window	0.943	0.956	2.0%	159	194 / 159	

Note: “PCAP/CSV” lists total windows per source; “Matched” is the intersection used for metrics (PCAP-only windows = 194 – 159 = 35).

TABLE V

COMPARATIVE EVALUATION OF CTGAD-X AGAINST BASELINE AND STATE-OF-THE-ART MODELS ON TEMPORAL GRAPH ANOMALY DETECTION

Model Category	Model Name	Architecture (Condensed)	Accuracy	F1 Score	ROC-AUC	AUPRC
Traditional Baselines	LOF	Local Outlier Factor on processed tabular features	0.9526	0.8038	0.9500	0.8000
	OCSVM	One-Class SVM on processed tabular features	0.9561	0.8524	0.9800	0.7800
	Isolation Forest	Tree Ensemble on processed tabular features	0.8995	0.6697	0.9200	0.4700
Unsupervised Temporal GNNs	GDN	GCNConv + TDM + NT-Xent + Centroid	0.8692	0.8787	0.9348	0.9462
	DySAT (Unsupervised)	GATv2 + Temporal Attention + Link Prediction	0.8482	0.8641	0.8630	0.8575
	TGAE	GATv2 + TDM + Temporal Attention + GRU Decoder	0.8045	0.8493	0.8709	0.8627
Proposed Framework	CTGAD-X (Proposed)	GATv2 + Temporal Attention + NT-Xent + k-NN	0.9909	0.9888	0.9908	0.9979

- 2) *Snapshot-Level Aggregation*: To obtain a fixed-size vector representation for each snapshot while retaining its global structural properties, we apply mean pooling across the node dimension of the structural embeddings:

$$h_t^{(\text{agg})} = \frac{1}{N} \sum_{v=1}^N h_v^{(t,l=1)} \quad (4)$$

This step transforms the sequence of node embedding matrices into a sequence of single graph embedding vectors, $\{h_t^{(\text{agg})}\}_{t=i}^{i+L-1}$, where each $h_t^{(\text{agg})} \in \mathbb{R}^{D_{\text{enc}}}$.

- 3) *Temporal Attention and Sequence Embedding*: Finally, to integrate information across the entire sequence and capture temporal dynamics, we feed the sequence of aggregated snapshot embeddings $\{h_t^{(\text{agg})}\}$ into a standard multi-head self-attention layer (denoted as TempAttn), complete with residual connections and layer normalization. The output corresponding to the *last* time step is used as the final, comprehensive embedding for the entire sequence

$$z_{S_i} = \left(\text{TempAttn}(\{h_t^{(\text{agg})}\}_{t=i}^{i+L-1}) \right)_{\text{last}} \quad (5)$$

This final vector $z_{S_i} \in \mathbb{R}^{D_{\text{enc}}}$ encapsulates the essential spatio-temporal characteristics of the sequence, making it suitable for subsequent contrastive learning and anomaly detection tasks.

- 2) *Contrastive Learning Objective*: We employed a contrastive self-supervised learning strategy to train the encoder

to produce discriminative embeddings without relying on explicit anomaly labels. This approach encourages embeddings of similar (normal) sequences to be close, while pushing dissimilar ones apart.

- *Training Data*: The contrastive learning phase is trained *exclusively* on sequences identified as “normal” (i.e., `normal_sequence_indices`). This focuses on learning to model the manifold of normal system behavior.
- *View Generation*: For each normal sequence S_i in a batch, two correlated views are generated implicitly. This is achieved by passing the same sequence S_i through the encoder and a subsequent projection head (a simple MLP mapping $D_{\text{enc}} \rightarrow D_{\text{proj}}$) twice. The inherent stochasticity of the Dropout layers ensures that the two resulting projected embeddings, z'_i and z'_j , are slightly different “views” of the same underlying data. This avoids the need for complex and domain-specific graph augmentations.
- *NT-Xent Loss*: The framework parameters (encoder θ_e and projection head θ_p) are optimized by minimizing the Normalized Temperature-scaled Cross-Entropy (NT-Xent) loss [7]. For a positive pair (z'_i, z'_j) from the same sequence and a batch of size B , the loss for view z'_i is:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\text{sim}(z'_i, z'_j)/\tau)}{\sum_{k=1, k \neq i}^{2B} \exp(\text{sim}(z'_i, z'_k)/\tau)} \quad (6)$$

where $\text{sim}(u, v) = u^\top v / (\|u\| \|v\|)$ is the cosine similarity, τ is the temperature hyperparameter, and the summation in the denominator is over all $2B$ embeddings in the

augmented batch. Minimizing this loss forces the representations of different views of the same normal sequence to be closer together, while pushing the representations of different normal sequences further apart.

3) *Why Contrastive Learning Supports Zero-Day Separability*: We provide a formal justification for using a contrastive objective (NT-Xent) over reconstruction-based methods for zero-day detection.

[Sufficient condition for k-NN separability] Let $d_k(z, \mathcal{R})$ be the average cosine distance from a test embedding z to its k nearest neighbors in the normal reference set $\mathcal{R}$. If the learned embedding satisfies:

- 1) *Normal Compactness*: $\forall z_n \sim p_{\text{norm}}, \text{ neighbors satisfy } \langle z_n, z_j \rangle \geq 1 - \epsilon$.
- 2) *OOD Angular Margin*: $\forall z_a \sim p_{\text{anom}}, \langle z_a, z \rangle \leq 1 - \gamma$ for all $z \in \mathcal{R}$.

Then, if $\gamma > \epsilon$, the anomaly score separates zero-day attacks with margin $\gamma - \epsilon$: $d_k(z_a) \geq \gamma > \epsilon \geq d_k(z_n)$.

Mechanism & Comparison: Reconstruction objectives minimize error $\mathbb{E}\|x - \hat{x}\|^2$ but do not explicitly constrain latent geometry. High-capacity decoders can often reconstruct anomalies (violating Condition 2) or map normal samples to dispersed codes (violating Condition 1). In contrast, NT-Xent directly optimizes these geometric properties: the alignment term minimizes ϵ (pulling views together), whereas the uniformity term maximizes γ (pushing distinct sequences apart).

Theoretical Role of Temperature (τ): The margin γ is structurally controlled by τ . As $\tau \rightarrow 0$, the loss approaches a hard-negative mining objective $\mathcal{L} \approx \max_k \text{sim}(z, z_k) - \text{sim}(z, z^+)$, focusing penalties on the most confusing negatives. This forces the model to define a strict boundary of normalcy, resulting in a tighter and more compact manifold. Conversely, a larger τ distributes gradients uniformly, smoothing the manifold and reducing the discriminative power. For zero-day detection, where distinct boundaries are critical, a lower τ (e.g., 0.1) is theoretically preferable.

4) *Implementation and Training Details*: CTGAD-X was implemented in PyTorch using PyTorch Geometric. Temporal graphs were constructed using the causality-aware flow slicing procedure in Algorithm 1 and organized into fixed-length sequences, as described in Section IV-A (Fig. 1). The encoder and projection head were trained in a self-supervised manner using only the normal sequences. For each sequence in a mini-batch, we generated two correlated views by performing two stochastic forward passes through the same network; dropout provided the required randomness without introducing domain-specific graph augmentations.

The model parameters were optimized by minimizing the NT-Xent objective in (6) using AdamW. We applied early stopping based on the NT-Xent validation loss to prevent overfitting and selected the best checkpoints. All architectural and optimization hyperparameters (e.g., sequence length, embedding dimensions, attention heads, learning rate, dropout, batch size, and temperature) are listed in Table III.

5) *Anomaly Scoring*: After contrastive training, the projection head was discarded, and the trained encoder was used for anomaly scoring. First, we established a reference manifold of

normal behavior by computing embeddings for all sequences in the normal training set, creating the reference set H_{normal} . For any given test sequence $\mathcal{S}_{\text{test}}$, its anomaly score S_{anomaly} is calculated as the average **cosine distance** to its k nearest neighbors (where $k = 15$) within this reference set:

$$S_{\text{anomaly}}(\mathcal{S}_{\text{test}}) = \frac{1}{k} \sum_{z_{\text{norm}} \in \mathcal{N}_k(z_{\text{test}}, H_{\text{normal}})} \left(1 - \frac{z_{\text{test}}^\top z_{\text{norm}}}{\|z_{\text{test}}\|_2 \|z_{\text{norm}}\|_2} \right) \quad (7)$$

where $\mathcal{N}_k(z_{\text{test}}, H_{\text{normal}})$ denotes the set of k nearest neighbors of the test embedding z_{test} in H_{normal} .

This scoring mechanism directly leverages the discriminative space learned during the contrastive training. Normal test sequences, which are similar to the training data, produce embeddings close to the H_{normal} manifold, resulting in a low anomaly score. Conversely, anomalous sequences, including unseen zero-day attacks, are expected to deviate significantly from the learned dynamics, yielding embeddings far from the manifold, and thus a high anomaly score. The k-NN cosine distance provides a robust, non-parametric measure of this semantic deviation.

Complexity: For N_{ref} reference embeddings of dimension D , brute-force k-NN scoring requires $\mathcal{O}(N_{\text{ref}}D)$ time per query and $\mathcal{O}(N_{\text{ref}}D)$ memory. For large N_{ref} , approximate nearest-neighbor indexing (e.g., FAISS-HNSW) or prototype compression ($M \ll N_{\text{ref}}$) can reduce the query cost to $\mathcal{O}(MD)$.

6) *Hyperparameter Optimization*: To ensure an unbiased evaluation, we performed a two-stage hyperparameter optimization process using only the training and validation splits of the data. First, core model parameters (e.g., learning rate, weight decay, dropout, attention heads, and dimensionality) were tuned via Bayesian Optimization, with the objective of minimizing the contrastive loss on the validation set ($\mathcal{G}_{\text{val}}$).

Second, after training the encoder with these optimal parameters, the k-NN hyperparameter (k) was selected. We evaluated the ROC-AUC on the validation set for a range of k values ($k \in \{5, 10, 15, 30, 50\}$) and chose the best-performing value, $k = 15$. All selected hyperparameters, including $k = 15$, were then frozen before the final evaluation on the held-out test set, thereby preventing data leakage.

7) *Knowledge Transfer and Fine-Tuning Strategy*: To evaluate the generalizability of our learned representations and enhance their practical applicability, we adopted a pre-training and fine-tuning paradigm. The CTGAD_Encoder is first pre-trained on a large-scale, diverse source dataset (e.g., UNSW-NB15) to learn a foundational manifold of normative network behavior. For adaptation to a new target domain, the model is initialized with these pre-trained weights and subsequently fine-tuned on the normal sequences of the target dataset. This approach leverages knowledge from a broad data domain to establish a robust starting point for learning domain-specific nuances of the data.

A key finding of this study is the remarkable efficiency of this adaptation process. We demonstrate that effective specialization is consistently achieved with minimal computational effort, typically requiring only approximately 20 fine-tuning epochs with a reduced learning rate to prevent catastrophic forgetting. This rapid convergence validates the robustness and domain

invariance of the pre-trained features, underscoring the practical value of our framework for real-world deployments, where extensive dataset-specific tuning is often infeasible. To ensure methodological consistency across benchmarks, we maintained a temporal window of $\Delta t = 30s$, adjusting only for smaller-scale datasets where a wider window is necessary to form meaningful graph structures.

C. Multi-Level Explainability Framework (XAI)

To ensure operational transparency, CTGAD-X integrates a multi-level interpretation module that triangulates anomaly causes from three perspectives, validated quantitatively via Fidelity+ and Stability metrics (Section VI-F).

1) *Structural & Temporal Attention (Qualitative Focus)*: We extracted the attention weights directly from the encoder to visualize the internal reasoning of the model. Structural attention weights ($\alpha^{(GAT)}$) are averaged across heads to identify high-weight edges within snapshots, highlighting influential interactions, such as scanning or lateral movement. Temporal attention weights ($\alpha^{(Temp)}$) reveal which historical snapshots contributed most to the final embedding, providing a temporal heat map of attack progression.

2) *Input Perturbation (Quantitative Attribution)*: To move beyond visualization, we quantified the causal impact of specific inputs using perturbations. For a flagged sequence S_i , we iteratively mask individual nodes or time steps with neutral values and measure the change in the k-NN anomaly score: $\Delta = S_{anomaly}(S_i) - S_{anomaly}(S_i^{masked})$. A large positive Δ confirms that the masked component was a primary driver of the anomaly, effectively answering the counterfactual: “Would this sequence still be anomalous without Node X ?”

3) *Global Permutation Importance (Strategy Validation)*: To validate the model’s high-level logic against domain expertise, we computed the Permutation Feature Importance on the test set. By randomly shuffling the values of a specific feature across the dataset and measuring the degradation in ROC-AUC, we quantified the model’s global reliance on that attribute. This ensures that the model learns robust behavioral concepts (e.g., traffic load and TTL variance) rather than overfitting to spurious identifiers.

V. EXPERIMENTAL EVALUATION STRATEGY

This section outlines the rigorous methodology used to validate the CTGAD-X framework. We detail the datasets, evaluation protocols, and key performance indicators used to measure the effectiveness, generalizability, and efficiency of the model.

A. Datasets and Evaluation Metrics

Our evaluation leverages several standard cybersecurity datasets, including UNSW-NB15 [9], CICIDS-2017 [10], ToN-IoT [11], SWaT [12], and SCVIC-APT-2021 [13]. All experiments were implemented in PyTorch and executed using an NVIDIA RTX 3090 GPU.

Across all evaluations, we measured performance using a comprehensive set of key metrics, such as Accuracy, F1-Score,

False Positive Rate (FPR), ROC-AUC, and Area Under the Precision-Recall Curve (AUPRC). To assess practical viability, we also report the inference Latency.

B. Evaluation Protocol

Our evaluation was structured into three distinct phases.

- 1) *Baseline Performance on UNSW-NB15*: We first conduct a direct comparison of CTGAD-X against a suite of traditional and graph-based baseline models (including OCSVM, Isolation Forest, Local Outlier Factor, DySAT, TGAE and GDN) on the UNSW-NB15 dataset using identical data splits and graph features.
- 2) *Generalization Across Benchmarks Through Transfer Learning*: To assess the robustness and adaptability of CTGAD-X, we adopt a transfer learning approach. The model was first trained on the UNSW-NB15 dataset to learn a foundational representation of normal behavior. It is then fine-tuned on each target benchmark dataset without reinitializing the encoder. This setup enabled the model to retain the core temporal and structural features while adapting to the unique characteristics of each dataset. The results demonstrate strong generalization across diverse network environments, validating the applicability of CTGAD-X in real-world heterogeneous settings.
- 3) *State-of-the-Art (SOTA) Comparison*: Finally, we contextualize our results by comparing the performance and methodology of CTGAD-X against other recently published SOTA methods in graph-based threat detection.

C. Comprehensive Zero-Day Generalization Analysis

To rigorously evaluate the zero-day detection capacity, we implemented a systematic leave-one-out protocol, where a single major attack category (e.g., ‘Exploits’, ‘Generic’) is completely held out during training. Consistent with our self-supervised methodology, the encoder was trained exclusively on normal data. We assessed performance using a dual-mode evaluation strategy: first, we measured *deployable performance* by applying a fixed baseline threshold calibrated on known traffic (simulating operational reality without access to zero-day labels); second, we calculated the *separability upper bound* using a post-hoc optimal threshold to quantify the intrinsic discriminative power of the learned embeddings against the novel threat morphology.

D. Implementation Details

The CTGAD-X framework was implemented using Python 3, PyTorch (v2.0+), and the PyTorch Geometric library. All experiments were conducted on a system equipped with an NVIDIA RTX 3090 GPU (24 GB VRAM), 10 virtual CPUs, and 60 GB RAM. The key hyperparameters used for training and evaluation are listed in Table III. Reference embeddings for k-NN scoring were consistently generated using only sequences identified as strictly normal.

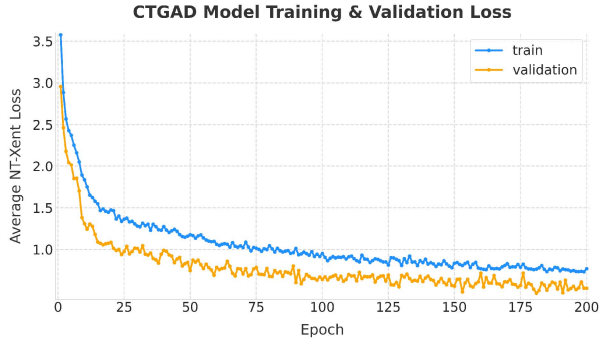


Fig. 2. CTGAD-X training and Validation loss (Average NT-Xent) over 200 epochs. The decreasing trend indicates successful learning of discriminative features for normal sequences.

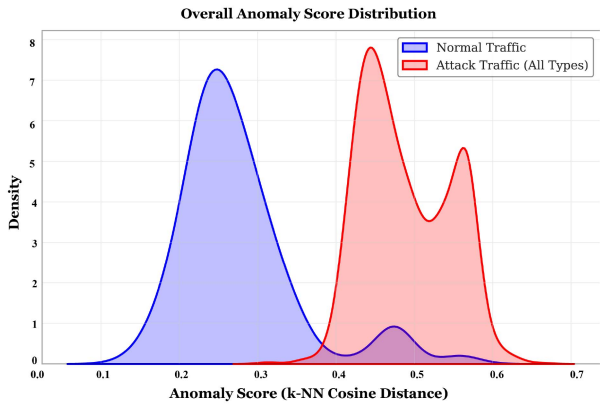


Fig. 3. Distribution of k-NN anomaly scores for Normal (blue) versus Attack (red) sequences. The distinct separation highlights the framework's discriminatory power.

VI. RESULTS AND ANALYSIS

This section presents the experimental results obtained from the evaluation of the CTGAD-X framework, focusing on its anomaly detection efficacy and ability to identify zero-day threats.

A. Anomaly Detection Performance

1) *Training Convergence*: The contrastive training process of the CTGAD-X framework demonstrated stable convergence. Fig. 2 shows the average NT-Xent loss per epoch on the training set. The loss decreased consistently, indicating that the encoder learned to differentiate between the augmented views of the same normal sequence and the views of different normal sequences. This process guides the model to learn a compact representation manifold for normal behavior, with the loss reaching a stable and low value after 200 epochs in our experiments.

2) *Overall Detection Efficacy*: The baseline performance of the framework was evaluated on the held-out test set (20% split, chronologically disjoint). As shown in Table IV(A), under rigorous causality-aware preprocessing, CTGAD-X achieves an ROC-AUC of 0.9908 and an F1-Score of 0.9888 (at optimal $k = 15$). This confirms that the high performance of the model is driven by genuine temporal pattern recognition rather than start-time artifacts. The comparative results in Table V further

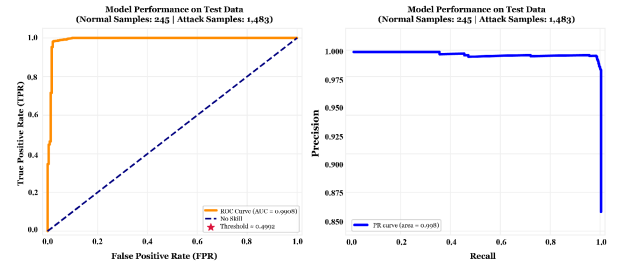


Fig. 4. Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves for the proposed CTGAD-X framework. The model achieves an ROC-AUC of 0.9908 and an AUPRC of 0.9979, demonstrating robust detection capabilities.

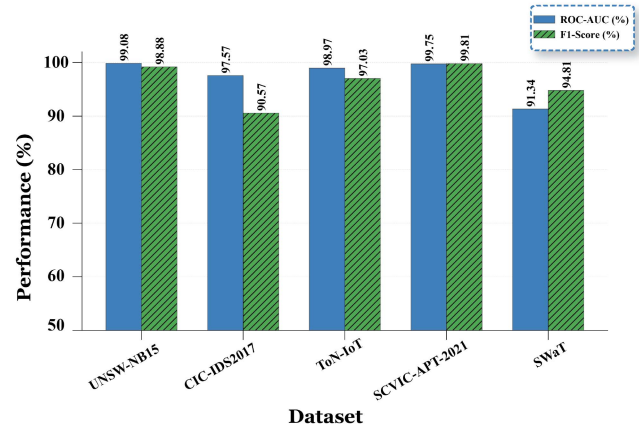


Fig. 5. Comparison of ROC-AUC and F1-Score (in %) across five benchmark cybersecurity datasets. The model demonstrates consistently high performance across UNSW-NB15, CICIDS-2017, SCVIC-APT-2021, and ToN-IoT, with a notable drop in ROC-AUC for SWaT, while F1-Score remains strong.

show that CTGAD-X outperforms the evaluated traditional baselines and unsupervised temporal GNN baselines under the same temporal-graph evaluation protocol.

The resulting anomaly-score distributions are shown in Fig. 3, where normal and attack sequences occupy distinct regions of the k-NN cosine-distance space, supporting the quantitative ROC/PR results. The discriminatory power was visually confirmed by the ROC and Precision-Recall curves in Fig. 4. The optimal operating point (Youden's J) was identified at a threshold of ≈ 0.5 , yielding a balanced trade-off between sensitivity and specificity. While we report this balanced configuration as the primary baseline, the system remains operationally flexible; security operations centers (SOCs) can adjust this threshold to prioritize either high-precision (minimizing false alarms) or high-recall (zero-day discovery), as explored in the Zero-Day Analysis (Section VI-C).

3) *Quantitative Evaluation: Generalization via Transfer Learning*: To validate the generalizability of our framework, we employed a transfer learning (TL) strategy. The CTGAD-X model was first trained on the comprehensive UNSW-NB15 dataset to learn a robust representation of general network behavior. Subsequently, this pre-trained model was fine-tuned on three diverse target datasets: CICIDS-2017 [10], ToN-IoT [11], SWaT [12] and SCVIC-APT-2021 [13]. This protocol directly

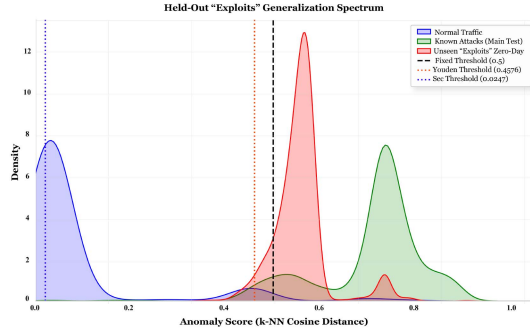
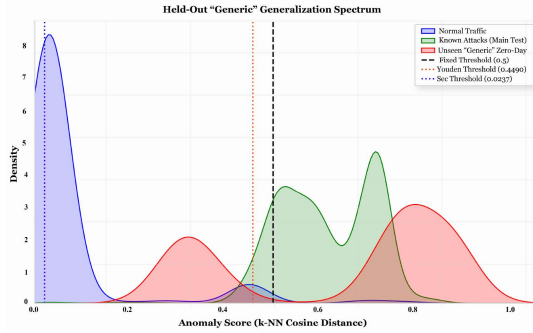
(a) Score distributions for held-out *Exploits*.(b) Score distributions for held-out *Generic*.

Fig. 6. Zero-day generalization under leave-one-out evaluation. The distributions of anomaly scores are shown for normal traffic, seen attack families, and the unseen held-out family. A fixed deployable threshold (0.5) controls false positives but can miss subtle zero-day attacks (notably, *Generic*). Label-calibrated thresholds (on normal + seen families only) illustrate the achievable sensitivity under different operating budgets; Recall-100 stresses the high-recall regime at the cost of very high false positives.

tests the model's ability to adapt its knowledge to new operational environments with minimal training.

As detailed in Fig. 5, the fine-tuned model maintained exceptional performance across all target datasets, achieving high ROC-AUC and F1-Score. This successful knowledge transfer validates the robustness of the learned representations and demonstrates the practical adaptability of the framework for deployment in diverse network security environments.

B. Parameter Sensitivity and Theoretical Validation

To validate the architectural choices, we analyzed the impact of key hyperparameters (Table VI):

- *Temperature (τ)*: Lower τ sharpens the contrastive boundary (hard-negative effect) and yields a more compact normal manifold. Across $\tau \in \{0.05, 0.1, 0.2, 0.5\}$ (Table VI(A)), performance is stable; we use $\tau = 0.1$ as a shared default across datasets for fair cross-dataset comparison.
- *Temporal Context (SEQ_LEN)*: Longer contexts marginally improve AUC but incur $\mathcal{O}(L^2)$ attention cost. We evaluated $L \in \{3, 5, 7, 10\}$ and selected $L = 5$ (Panel B) as the best accuracy–efficiency trade-off.
- *Scoring Robustness (k , Metric)*: Performance is stable across $k \in \{5, 15, 30, 50\}$ (Panel C) and comparable between euclidean and Cosine metrics, confirming detection

TABLE VI
SENSITIVITY ANALYSIS OF KEY HYPERPARAMETERS. THE MODEL DEMONSTRATES ROBUSTNESS ACROSS TEMPERATURE (τ), SEQUENCE LENGTH, AND NEIGHBORHOOD SIZE (k) AT CAUSALITY-AWARE SLICING AT 30 s WINDOW

Parameter	Value	ROC-AUC	AUPRC	F1-Score
A. Contrastive Temperature (τ)				
	0.05	0.9987	0.9998	0.9899
	0.1 (Default)	0.9908	0.9979	0.9888
	0.2	0.9772	0.9955	0.9879
	0.5	0.9938	0.9987	0.9879
B. Sequence Length (SEQ_LEN)				
	3	0.9560	0.9880	0.9858
	5 (Default)	0.9908	0.9979	0.9888
	7	0.9889	0.9981	0.9892
	10	0.9985	0.9998	0.9919
C. Neighborhood Size (k)				
	5	0.9691	0.9927	0.9879
	15 (Default)	0.9908	0.9979	0.9888
	30	0.9775	0.9953	0.9879
	50	0.9799	0.9961	0.9879

is driven by manifold quality rather than fragile local assumptions.

- *Efficiency*: The encoder (16,320 parameters, $D = 60$) achieves 3.94 ms inference latency, suitable for edge deployment.
- *Scoring Metric (Cosine vs. Euclidean)*: Under causality-aware slicing with 30 s windows, euclidean k-NN scoring achieves ROC-AUC = 0.9890 while cosine k-NN scoring achieves ROC-AUC = 0.9908; therefore, we keep cosine as the default scoring metric.

C. Empirical Analysis of Zero-Day Generalization

We evaluated zero-day detection using a leave-one-out protocol on UNSW-NB15, where one major attack category was fully held out during model development. We considered two held-out categories, *Exploits* and *Generic*, and trained a separate CTGAD-X model for each protocol. The anomaly score is computed using the k-NN distance in the learned embedding space. The held-out family is excluded from the threshold selection and is used only for final reporting. Fig. 6 visualizes the score distributions and threshold locations for the held-out Exploits and Generic protocols, illustrating how conservative and high-sensitivity operating points trade false alarms for zero-day recall.

To clarify the operational meaning of thresholding, we report three operating points. All label-calibrated thresholds were selected using only the calibration stream containing *normal traffic* and *seen attack families*, excluding the held-out family. The fixed threshold does not require calibration. Each selected threshold was applied once to the corresponding held-out test split.

1) Mode 1: Deployable Performance (No Calibration):

- *Fixed (Deployable)*: We use a fixed threshold of 0.5 without tuning. This is a conservative label-free operating point that

TABLE VII

ZERO-DAY GENERALIZATION WITH 95% CONFIDENCE INTERVALS (WILSON). THE DETECTION RATE WAS MEASURED ON THE HELD-OUT ZERO-DAY TEST SPLIT. THE MAIN-STREAM FPR IS MEASURED ON THE MAIN TEST STREAM (NORMAL + SEEN FAMILIES). THE FIXED THRESHOLD IS LABEL-FREE AND DIRECTLY DEPLOYABLE; YODEN-J AND RECALL-100 ARE CALIBRATED USING LABELS FROM THE CALIBRATION STREAM (NORMAL + SEEN FAMILIES), EXCLUDING THE HELD-OUT FAMILY

Held-Out Threat	Operating Point	Threshold	Main Test FPR	Detection Rate (Count)	95% Confidence Interval
Exploits	Fixed (Deployable)	0.5000	1.63%	88.81% (1413/1591)	[87.17, 90.27]
	Youden-J (Calibrated)	0.4576	1.83%	96.86% (1541/1591)	[95.88, 97.61]
	Recall-100 (High Sens.)	0.0247	71.54%	100.00% (1591/1591)	[99.76, 100.00]
Generic	Fixed (Deployable)	0.5000	1.23%	65.46% (995/1520)	[63.03, 67.81]
	Youden-J (Calibrated)	0.4490	1.44%	65.66% (998/1520)	[63.23, 68.00]
	Recall-100 (High Sens.)	0.0237	75.00%	100.00% (1520/1520)	[99.75, 100.00]

can be deployed as is. As shown in Table VII, it yields low main-stream false positive rates (1.63% for the *Exploits* protocol and 1.23% for the *Generic* protocol), but it can miss subtle zero-day behaviors, particularly for *Generic* (65.46% detection).

2) *Mode 2: Label-Calibrated Operating Points (Seen Families Only)*: These operating points use labels from the calibration stream (normal + seen families) to select thresholds. They remain valid for deployment in environments where historical labels for known threats exist; however, they do *not* use any labels from the held-out family.

- *Youden-J (Calibrated)*: We select the threshold that maximizes Youden’s J statistic on the calibration stream (normal + seen families). This improves held-out *Exploits* detection (96.86%) at low main-stream false-positive rates, while *Generic* shows only marginal improvement (65.66%), indicating weaker separability for that family under strict alert budgets.
- *Recall-100 (High-Sensitivity Stress Point)*: We select the smallest threshold that achieves 100% recall on the calibration stream. This setting attains 100% detection on the held-out family in both protocols, but at the cost of extremely high main-stream false positive rates (71.54% and 75.00%). This point was included to quantify the recall–false-alarm trade-off rather than as a practical SOC configuration.

SOC implication: A SOC cannot tune thresholds using labels from the novel (held-out) family. In practice, operators would set thresholds using benign data and historically observed threat labels, guided by an alert volume or false-positive budget. Table VII shows that CTGAD-X exposes a controllable trade-off: a conservative deployable threshold minimizes false alarms but can miss stealthier zero-days, whereas label-calibrated operating points quantify the achievable sensitivity under explicit operational budgets.

D. Practical Deployment and Edge Feasibility

To address the operational requirements of modern NIDS, we assessed the deployment feasibility of CTGAD-X in practical workflows, particularly in edge and federated environments.

Operational Workflow: CTGAD-X integrates into SOC workflows as a semantic anomaly filter. A reference bank of normal

TABLE VIII
STATISTICAL RELIABILITY ANALYSIS (MEAN $\pm$ STD DEV ACROSS 3 INDEPENDENT TRAINING SEEDS)

Metric	Mean $\pm$ Std Dev
ROC-AUC	0.9883 $\pm$ 0.0254
AUPRC	0.9837 $\pm$ 0.0139
F1-Score	0.9909 $\pm$ 0.0064
Accuracy	0.9844 $\pm$ 0.0110
Precision	0.9904 $\pm$ 0.0067
Recall	0.9915 $\pm$ 0.0068

embeddings is precomputed offline. At runtime, live traffic sequences are encoded and scored using the k-NN distance to this bank. This “retrieval-based” detection allows for agile updates: operators can incrementally update the definition of “normalcy” by appending verified embeddings to the reference bank without expensive model retraining.

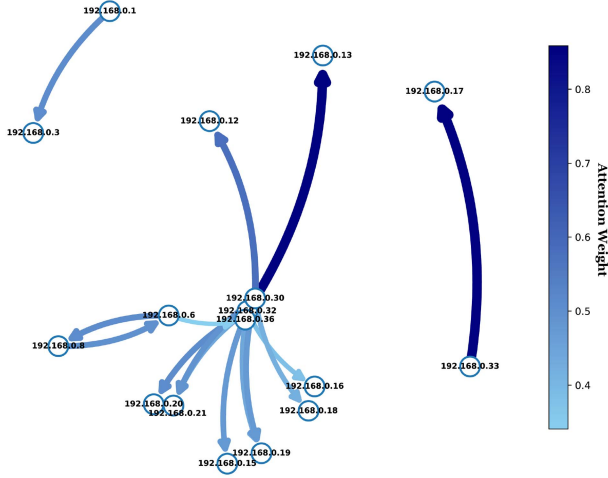
Edge and Federated Suitability: With a minimal footprint of 16,320 parameters and an inference latency of 3.94 ms per sequence, the encoder is highly suitable for resource-constrained edge gateways. A potential constraint for large-scale deployments is the memory growth of the k-NN reference set. This can be effectively mitigated using approximate nearest neighbor (ANN) indexing (e.g., FAISS) or prototype compression (storing only cluster centroids). These optimizations reduce query complexity and memory usage, serving as key enablers for future federated learning architectures where lightweight and privacy-preserving model distribution is paramount.

E. Statistical Reliability and Reproducibility

To address concerns regarding the variability of contrastive learning due to stochastic initialization and optimization, we evaluated the statistical reliability of CTGAD-X. We retrained the complete framework from scratch using three independent random seeds (42, 43, 44) while keeping all preprocessing and hyperparameters fixed.

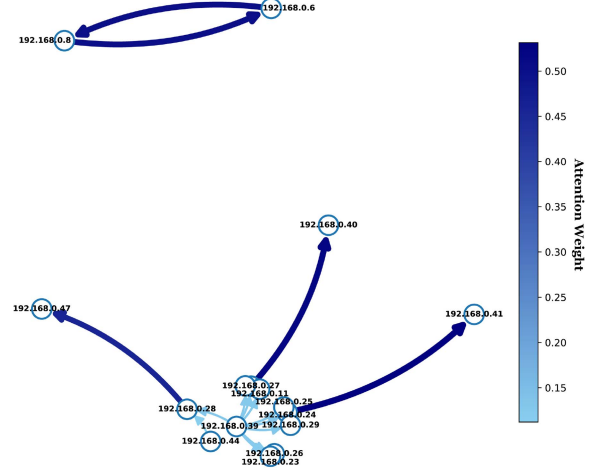
Table VIII reports the mean and standard deviation for key performance metrics. The model exhibited high stability, with F1-Scores consistently exceeding 0.98 and a low standard deviation of ± 0.0064 . Although the ROC-AUC shows slight fluctuation (± 0.0254) owing to the geometry of the embedding space varying slightly between runs, the overall detection capability

**Structural Attention Map
Seq 1074 (Attack)**



(a) Attack Sequence 1074

**Structural Attention Map
Seq 257 (Normal)**



(b) Normal Sequence 257

Fig. 7. Structural attention visualization for the last snapshot (T4). (a) In the attack (Seq 1074), attention concentrates on aggressive scanning-like behaviors from specific source nodes. (b) In the normal sequence (Seq 257), attention is distributed across routine, balanced internal communications.

remains robust. These results confirm that the reported performance is reproducible and not an artifact of the specific random seed.

F. Explainability Evaluation

In addition to detection accuracy, operational adoption requires transparent reasoning. This section evaluates the CTGAD-XAI module, which integrates instance-level attention analysis with input perturbation to validate the decision logic of the model.

1) *Instance-Level Explanation. Identifying Anomaly Drivers:* To interpret specific alerts, the XAI module provides two complementary perspectives: the internal structural focus (attention) of the model and the quantitative impact of inputs on the anomaly score (perturbation).

For the high-scoring attack sequence (Seq 1074), the structural attention correctly isolates aggressive communication patterns originating from specific source nodes, such as 192.168.0.33 and 192.168.0.32. As shown in Fig. 7(a), the model assigns exceptionally high attention weights (> 0.85) to these edges, flagging them as the primary anomaly drivers. In contrast, for the normal sequence (Seq 257), the attention map highlights sparse, balanced interactions characteristic of routine traffic, such as the bidirectional communication between 192.168.0.6 and 192.168.0.8 (Fig. 7(b)).

Input perturbation analysis quantitatively validates these insights. For the attack sequence, masking the features of the identified aggressor nodes (e.g., 192.168.0.33) resulted in a significant decrease in the anomaly score, confirming their activity as the cause of the alert. Conversely, perturbing key

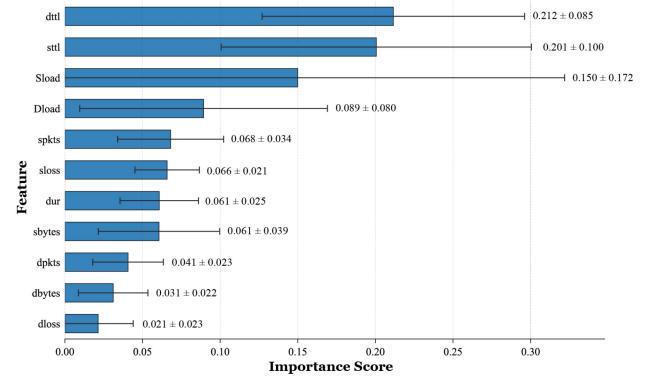


Fig. 8. Global permutation feature importance. The bars represent the decrease in ROC-AUC upon feature permutation (Mean ± Std). The model exhibits a strong reliance on Time-To-Live (TTL) and Load metrics.

nodes in the normal sequence disrupted the learned benign patterns, verifying that the model correctly relied on these routine interactions to establish a baseline of normalcy.

2) *Global Explanation. Validating the Learned Strategy:* To validate that our framework learns a logically sound detection strategy, we performed a global permutation feature importance analysis. This technique quantifies model reliance on each feature by measuring the degradation in test set ROC-AUC when that feature's values are randomly shuffled. The results in Fig. 8 reveals that the model prioritizes nuanced network-layer metadata and traffic intensity over simple duration statistics.

The analysis unequivocally identified time-to-live (TTL) features as the most critical factors. Permuting the destination TTL (dttl) and source TTL (sttl) caused the most significant performance drop, reducing the ROC-AUC by 0.212 and 0.201, respectively. This dependence suggests that the model

has learned to perform soft fingerprinting of network origins and topological distances (hop counts) to identify anomalies.

Beyond the TTL, the model places substantial weight on traffic throughput, with Source Load (Sload) emerging as the third most influential feature (importance score of 0.150), followed by Destination Load (Dload). In contrast, basic volumetric features such as duration (dur) and packet counts (spkts) played a secondary role (scores < 0.07). This validates that CTGAD-X learns a sophisticated detection logic based on behavioral semantics rather than relying solely on easily manipulable volumetric thresholds.

3) *Quantitative Validation of Explainability*: To address the need for a rigorous evaluation of the proposed multi-level XAI framework, we conducted a comprehensive quantitative assessment focusing on fidelity, stability, and consistency.

1) *Faithfulness (Fidelity+)*: We measured whether the features identified as important truly drive the model’s decision. Masking the Top-3 features yielded a relative score drop of 36.08% ($\pm 18.91\%$), confirming that the explanation pinpoints the actual anomaly drivers.

2) *Robustness to Noise*: Under Gaussian noise ($\sigma = 0.02$), the explanations achieved a Top-3 Jaccard overlap of 0.9850 (± 0.0105), indicating high stability against input fluctuations.

3) *Cross-Seed Stability*: Global feature importance rankings across three distinct training seeds (42, 43, 44) showed perfect agreement on the Top-3 features (Overlap = 1.0), confirming that the explanations reflect the learned manifold rather than initialization artifacts.

4) *Sample Consistency*: To evaluate whether the explanation strategy is consistent across the dataset, we computed the pairwise Spearman rank correlation of explanation vectors across 300 test sequences. The mean correlation was 0.5552 (± 0.2912). This moderate agreement with substantial variance is expected, as it reflects the model adapting its focus to the heterogeneous nature of different traffic behaviors rather than applying a static importance vector.

G. Robustness to Adversarial Evasion and Label Noise

To evaluate the dependability of CTGAD-X under hostile conditions, we conducted stress tests focusing on adversarial evasion and training-data contamination. The results using causality-corrected data are presented in Table IX.

1) *Adversarial TTL Evasion (Distribution Matching)*: We simulated a sophisticated adversary that attempted to evade detection by mimicking benign traffic patterns.

- *Mechanism*: We modified only the Time-To-Live features (sttl, dttl) of attack snapshots in the test set. To ensure a realistic spoof, we employed *distribution matching*: the attack TTL values were quantile-mapped to the distribution of the benign training TTLs. This removes the obvious “attack signature” while preserving valid statistical properties.
- *Result*: Under this stressor, the ROC-AUC remained relatively high (0.9654), indicating that the model still correctly ranks anomalies higher than normal traffic based on structural and flow-rate features. However, the F1-Score dropped to 0.8303, suggesting that while separation exists,

TABLE IX
ROBUSTNESS ANALYSIS UNDER ADVERSARIAL STRESSORS
(CAUSALITY-CORRECTED DATA). WE COMPARE THE CLEAN BASELINE
AGAINST ADVERSARIAL TTL SPOOFING (TEST-TIME EVASION) AND RANDOM
LABEL NOISE (TRAINING-TIME POISONING).

Scenario	ROC-AUC	PR-AUC	F1-Score
Clean Baseline	0.9908	0.9979	0.9888
Adversarial Evasion (Test-Time)			
TTL Spoofing (Dist. Match)	0.9654	0.9927	0.8303
Data Poisoning (Training-Time)			
Label Noise (1%)	0.9418	0.9575	0.9722
Label Noise (5%)	0.8519	0.9518	0.8624

the optimal decision boundary becomes less distinct when TTL signals are neutralized.

2) *Training Data Contamination (Label Noise)*: To assess resilience against data poisoning, we injected label noise by randomly replacing a fraction of the benign training snapshots with attack snapshots (altering both the adjacency and features).

- *1% Contamination*: The model showed strong resilience, maintaining a ROC-AUC of 0.9418 and F1 of 0.9722. This suggests that the contrastive objective is robust to minor outliers in the training data distribution.
- *5% Contamination*: Performance degraded more significantly (ROC-AUC 0.8519), confirming that maintaining training data hygiene is critical for optimal model performance.

Feature Ablation Study: To explicitly address concerns regarding potential overfitting to specific artifacts (e.g., Time-To-Live features), we performed a controlled ablation study. We retrained and evaluated the model with all TTL-related features removed from both the training and testing sets. The model retained strong discriminative power, achieving an ROC-AUC of 0.9368 (compared to the baseline of 0.9908). This modest degradation confirms that while TTL features provide a strong signal, the CTGAD-X framework successfully learns robust alternative representations from flow dynamics and structural patterns and does not rely solely on dataset-specific artifacts.

H. Cross-Dataset Validation of Explanations

To verify that the proposed CTGAD-XAI explanations are not specific to UNSW-NB15, we applied an identical explanation protocol to additional benchmark datasets used in our transfer learning evaluation. We report (i) global permutation feature importance as the ROC-AUC drop after permuting a feature across nodes and time steps and (ii) instance-level drivers via input perturbation for top-ranked anomalous sequences. Table X summarizes the results.

The cross-dataset analysis confirmed that the explanations aligned with the specific threat semantics of each domain. While UNSW-NB15 detection is driven by TTL artifacts, SCVIC-APT explanations correctly prioritize backward-volume and duration statistics (indicative of C2 exfiltration), and CICIDS2017 explanations focus on packet-length variance (a key differentiator for payload-based attacks). This demonstrates that the XAI module

TABLE X
CROSS-DATASET VALIDATION OF CTGAD-X EXPLANATIONS. THE SAME XAI PROTOCOL IS APPLIED ACROSS DATASETS: GLOBAL PERMUTATION FEATURE IMPORTANCE (ROC-AUC DROP) AND INSTANCE-LEVEL PERTURBATION DRIVERS FOR TOP ANOMALIES.

Dataset	Window	#Nodes	#Feat	Test ROC-AUC / F1	Global top drivers (largest ROC-AUC drop) and representative instance drivers
UNSW-NB15 [9],	30s	48	11	0.9908 / 0.9888	Global: TTL-derived features dominate (e.g., dttl_mean, sttl_sum). Instance: attention and perturbation highlight attacker-linked nodes and early time steps as primary contributors to anomalies.
SCVIC-APT [13]	5s	54	24	0.9975 / 0.9981	Global: Total Length of Bwd Packet_mean (largest drop), Flow Duration_mean, and Total Bwd packets_mean. Instance: The top anomalies are driven by the same backward-volume and duration statistics.
CICIDS2017 [10]	30s	9,015	27	0.9757 / 0.9057	Global: Packet Length Variance (largest drop), followed by Fwd Packet Length Max, Bwd Packet Length Mean, Packet Length Std, Max Packet Length. Instance: the top anomalies are consistently attributed to packet-length dispersion and extrema.

TABLE XI
COMPARISON OF CTGAD-X WITH STATE-OF-THE-ART GRAPH-BASED INTRUSION DETECTION MODELS

Author	Dataset(s)	Method Type	Key Results
Caville et al.(Anomal-E) [24]	NF-UNSW-NB15	Flow graphs (edge features), Self-supervised	F1-Score 92.35%, Acc 98.66%, DR 98.77%;
Wu et al.(Tcg-ids) [28]	UNSW-NB15	Temporal flow graphs, Self + Supervised	F1-Score 91.38%, Acc 91.48% , FPR 16.71%
Van Langendonck et al.(Ppt-GNN) [23]	NF-UNSW-NB15	Flow-IP hybrid line graphs, Self + Fine-tuned	F1: 97.77%
Jahin et al.(Cagn-GAT) [25]	CICIDS2017	Feature similarity graph, Supervised	Acc: 99.5%, F1: 0.9836, AUC: 0.87; Latency: 2.2–5.5s
Tran and Park(Fn-GNN) [31]	UNSW-NB15	Flow-based graph, Supervised	F1-Score: 0.9976 Acc 0.9865; Prediction time: 0.16s
Chen et al.(Dgmn) [17]	SWaT	Correlation subgraph, Unsupervised	F1: 96–98%
Proposed (CTGAD-X)	UNSW-NB15	Temporal behavior graphs, Self-supervised	ROC-AUC: 0.9908, Acc: 0.9909, F1: 0.9888

Note: Cross-method comparisons are approximate and intended to show relative trends (not a strict head-to-head benchmark).

generalizes effectively, identifying domain-relevant anomaly drivers without dataset-specific tuning.

Limitations and Future Work: While the cross-dataset validation demonstrates adaptability, explanation generalization remains a challenge. If the model relies on dataset-specific artifacts (e.g., TTLs in UNSW-NB15), the explanations will faithfully reflect this bias. Furthermore, while our quantitative metrics (Fidelity+, Stability) provide statistical validation, this study did not include a *human-in-the-loop* evaluation by security analysts. Future work will focus on qualitative user studies to assess the operational utility of these explanations and explore generating *counterfactual explanations* to prescribe precise remediation actions.

I. Computational Complexity and Inference Performance

To evaluate the operational feasibility of CTGAD-X, we analyzed its theoretical complexity and empirical runtime.

Complexity Analysis: The computational cost is primarily driven by the GATv2 and Temporal Attention layers. For a sequence of length L (fixed at 5), graph snapshots with $|V|$ nodes and $|E|$ edges, and a hidden dimension D :

- **Graph Encoder:** The structural attention mechanism scales linearly with the number of edges, $\mathcal{O}(L \cdot (|V| + |E|) \cdot D)$, making it efficient for sparse network graphs.
- **Temporal Attention:** The self-attention mechanism scales quadratically with sequence length, $\mathcal{O}(L^2 \cdot D)$. Given the short window ($L = 5$), this overhead becomes negligible.

- **Anomaly Scoring:** The k-NN search scales with the size of the reference set N_{ref} , i.e., $\mathcal{O}(N_{ref} \cdot D)$. In deployment, this can be accelerated to $\mathcal{O}(\log N_{ref})$ using indexing structures, such as FAISS.

k-NN scoring latency: The reported 3.94 ms reflects *GPU encoder* inference only; k-NN anomaly scoring is executed separately (CPU in our current implementation), and its per-query timing and memory footprint are reported in Section IV-B5. The end-to-end latency depends on the deployment pipeline (e.g., device transfer and whether k-NN is served on CPU or GPU).

Empirical Benchmarks: The resulting model is extremely lightweight, containing only **16,320 parameters**. We benchmarked the inference on a single NVIDIA GeForce RTX 3090 GPU (batch size = 1) to simulate real-time processing. After a warm-up phase (100 batches), the model achieved an average latency of 3.94 ms per sequence, corresponding to a throughput of approximately 253 sequences/s. This confirms the suitability of the framework for high-speed, near-real-time intrusion detection.

J. Comparison With State-of-the-Art Methods

To situate CTGAD-X within the current research landscape, we compared it with two distinct categories of state-of-the-art models.

First, we acknowledge the leading supervised frameworks designed for threat detection. In particular, ANUBIS by Anjum et al. [21] achieved a state-of-the-art F1-score of 0.996 using

a Bayesian Neural Network. Similarly, our previous adaptive framework [30] achieved an F1-score of 0.999 on UNSW-NB15 by employing a supervised stacking ensemble (fusing Recurrent Autoencoders, Isolation Forests, and a Random Forest meta-classifier) refined by Deep Q-Learning. While highly accurate, such architectures introduce significant ensemble complexity and computational overhead for training the reinforcement learning agent and critically depend on attack labels.

Given this key distinction, a direct performance benchmark of CTGAD-X is more meaningful against other self-supervised and unsupervised systems that share the goal of anomaly-based detection. A detailed comparison is presented in Table XI. The results show that our framework significantly outperforms existing frameworks. CTGAD-X achieved a superior ROC-AUC of 0.9908 and an exceptionally low false-positive rate (FPR) of 0.0204, highlighting the efficacy of its contrastive learning strategy on temporal graphs for dependable and secure threat detection.

VII. CONCLUSION

This study introduced CTGAD-X, an explainable self-supervised framework for zero-day threat detection on temporal graphs. CTGAD-X learns a discriminative embedding of normal behavior from benign traffic only via contrastive learning and flags anomalies through distance-based deviation from a normal reference bank. To enforce temporal correctness, we introduced causality-aware overlap-based flow slicing and validated window-level attribution against packet timestamps extracted from raw packet-capture (PCAP) traces, achieving a strong agreement (Pearson $r = 0.943$; median relative error 3.2% for bytes and 2.0% for packets). On UNSW-NB15 under causality-aware slicing, CTGAD-X achieves ROC-AUC = 0.9908 and F1-score = 0.9888. Transfer evaluation across additional benchmarks further indicates strong generalization with minimal adaptation, with ROC-AUC ranging from 0.9134 to 0.9975 and F1-score from 0.9057 to 0.9981. The leave-one-out zero-day evaluation highlights operational trade-offs across deployable thresholds: detection for held-out *Exploits* rises from 88.81% (conservative) to 100.00% (high-sensitivity), and for held-out *Generic* from 65.46% to 100.00%, at the cost of a higher false-positive rate (70-75%). The conservative operating point exposes a visibility gap for stealthier families with strict alert budgets. CTGAD-X integrates multi-level explainability (attention attribution, input perturbation, and global permutation importance) with quantitative faithfulness (Fidelity⁺: 36% mean anomaly-score drop under Top-3 feature masking) and explanation stability (Top-3 Jaccard = 0.985). Dependability analysis further shows that adversarial TTL spoofing preserves ranking quality (ROC-AUC = 0.9654) despite reduced boundary sharpness (F1 = 0.8303), whereas training contamination degrades performance progressively (F1 = 0.9722 at 1% versus 0.8624 at 5%).

Despite these results, limitations remain: the evaluation uses public benchmark datasets with observable flow-level metadata, whereas many real deployments operate over encrypted traffic, where payload semantics are unavailable and side-channel

distributions may shift. Future work will therefore (i) evaluate and adapt CTGAD-X under encrypted-traffic settings using robust representations of metadata, timing, and behavioral side channels; (ii) extend detection toward response by coupling the model with reinforcement-learning-based alert triage under operator constraints; and (iii) investigate privacy-preserving federated and edge deployment to enable cross-organization threat intelligence without exposing sensitive traffic data.

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